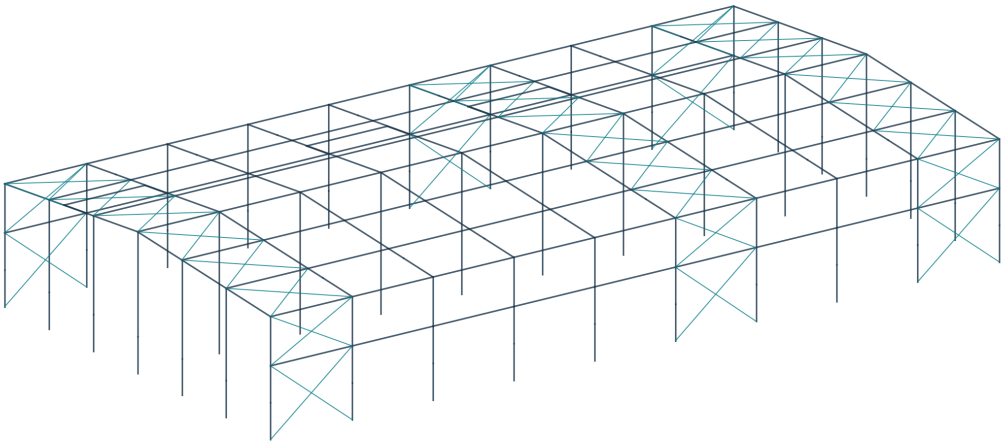
A
STRUCTURAL ENGINEERING TRAINING REPORT

ON

ANALYSIS, DESIGN AND DETAILING OF A PRE-ENGINEERED STEEL BUILDING

Proposed PEB Industrial Building, Chennai



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Institution	Demo College of Engineering
Training organization	Demo Structural Consultants
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Training period	Four weeks — dates to be entered
Report issue	21 October 2026 | Revision D1

36 m × 72.2 m • Two 18 m roof spans • Four-week study

Demonstration personal details. Prepared from the supplied project records.
Study drawings and illustrative calculations are identified in the report.

ABSTRACT

Structural engineering training connects classroom principles with the practical interpretation of project drawings, loading assumptions, analysis models and design output. In a pre-engineered steel building, the geometry of the primary frames, the restraint provided by secondary steel and the behaviour of the connections must be considered together. This report presents a four-week study of these aspects using the project records supplied for a proposed PEB industrial building in Chennai.

The building has a plan dimension of 36 m × 72.2 m, a roof slope of 1:12, an analytical eave height of 10 m and a ridge height of 11.5 m. Ridge columns divide the roof into two 18 m horizontal spans. The structural system comprises tapered built-up steel columns and rafters, longitudinal hollow-section members, tension-only rod bracing, cold-formed secondary members and profiled roof and wall cladding. The supplied STAAD.Pro model contains 180 joints, 314 members and 38 pinned supports.

The first week covers preliminary project information, types of structures, design-load calculations and load cases. The second week explains the structural model, the analysis of members, shear force and bending moment diagrams, member forces and displacement checks. The third week addresses proportioning, member dimensions, strength and stability checks for columns, rafters, secondary steel and bracing, together with connection-design requirements. The fourth week covers general detailing notes, dimensioned arrangement drawings, member profiles and connection concepts.

The study is based on the supplied 68-page training PDF and R1_PEB.ANL analysis output. Calculations and recorded results are distinguished from illustrative teaching examples. Review of the analysis identified a load-combination discrepancy, two failed member checks and unresolved analysis/design warnings; these are documented in the technical chapters. No revised STAAD run or final connection/foundation design is claimed. The report provides a structured academic account of the design process, with demonstration personal details and drawings intended for training use.

TABLE OF CONTENTS

ABSTRACT **2**

CHAPTER 1: INTRODUCTION **5**

1.1 General 5
1.2 About the Project 5
1.3 Location and Site Details 5
1.4 Type of Structure 5
1.5 Objectives and Scope of the Training 6
1.6 Methodology 6
1.7 Organization of the Report 6
1.8 Documentation and Evidence Basis 6

CHAPTER 2: DETAILED PROJECT DESCRIPTION **7**

2.1 Preliminary Information and Structural Systems 7
2.1.1 Types of structures considered during training 7

CHAPTER 3: STRUCTURAL ENGINEERING TRAINING – 1st WEEK **8**

3.1 Dead Load and Imposed Load Calculations 8
3.1.1 Dead load (DL) 8
3.1.2 Roof imposed load and gutter water 8
3.2 Wind and Earthquake Load Calculations 9
3.2.1 Wind calculation reproduced from the training PDF 9
3.2.2 Earthquake calculation reproduced from the analysis 9
3.3 Load Cases and Combination Register 10
3.3.1 Input checks identified during the study 10

CHAPTER 4: STRUCTURAL ENGINEERING TRAINING – 2nd WEEK **11**

4.1 Structural Model and Analysis Procedure 11
4.1.1 Analysis sequence and interpretation 11
4.2 Shear Force and Bending Moment Diagrams 12
4.2.1 Example A — simply supported reference beam 12
4.2.2 Example B — canopy reference cantilever 12
4.3 Recorded Forces in Representative Members 13
4.3.1 Engineering interpretation 13
4.4 Displacements, Equilibrium and Model Review 14
4.4.1 Primary-case displacement summary 14
4.4.2 Serviceability interpretation 14
4.4.3 Equilibrium and warnings 14

TABLE OF CONTENTS (CONTINUED)

CHAPTER 5: STRUCTURAL ENGINEERING TRAINING – 3rd WEEK **15**

5.1 Proportioning and Member-Size Schedule 15
5.1.1 Reading the tapered-section input 15
5.2 Design of Columns and Gable Posts 16
5.2.1 Column design procedure 16
5.2.2 Worked interpretation — ridge column 154 16
5.2.3 Exterior columns and gable posts 16
5.3 Design of Rafters and Canopy Beams 17
5.3.1 Rafter design principles 17
5.3.2 Worked interpretation — ridge-side rafter 39 17
5.3.3 Canopies and end-frame rafters 17
5.4 Secondary Steel and Bracing Design 18
5.4.1 Illustrative roof-purlin calculation 18
5.4.2 Illustrative gross-yielding resistance of a Ø25 brace 18
5.5 Design Coverage, Connections and Quantities 19
5.5.1 Coverage of the supplied design output 19
5.5.2 Connection design method 19
5.5.3 Steel quantity reported by the model 19

CHAPTER 6: STRUCTURAL ENGINEERING TRAINING – 4th WEEK **20**

6.1 General Notes for Design and Detailing 20
6.1.1 Drawing register 20
6.2 General Arrangement and Frame Elevation 21
6.3 Member Profiles and Section Details 22
6.4 Connection Details and Drawing Completion 23

CHAPTER 7: CONCLUSION **24**

7.1 Learning Outcomes, Conclusions and References 24
7.1.1 Weekly learning outcomes 24
7.1.2 Conclusion and remaining engineering work 24
7.1.3 Source register and references 24

CHAPTER 1: INTRODUCTION

1.1 General

Structural engineering deals with the analysis and design of systems that transfer loads safely to the ground while providing suitable stiffness, stability and durability. For steel buildings, member resistance must be considered together with local buckling, overall stability, connection behaviour and the restraint available during construction and service. The design process therefore extends from understanding the project requirements to coordinating the analytical model with the drawings.

A pre-engineered building uses an engineered combination of primary frames, secondary members and cladding. Tapered built-up sections allow the distribution of steel to follow the structural demand. Repeated framing arrangements support efficient fabrication and erection, but the benefits depend on correctly defined loads, reliable connections and adequate bracing. This training report follows that process through the supplied PEB case study.

1.2 About the Project

The case study is a proposed industrial PEB with a 36 m transverse width and 72.2 m longitudinal length, giving a plan area of 2,599.2 m². Ten transverse frame lines form nine longitudinal bays. The two end bays are 8.1 m long and the seven intermediate bays are 8.0 m long. The analytical eaves are at 10.0 m and the ridge is at 11.5 m, consistent with a roof rise of 1.5 m and a slope of 1:12.

The model includes interior ridge columns and gable-end posts. Accordingly, the 36 m building width is not an unobstructed clear span: the roof is supported over two 18 m horizontal spans. The source input sheet states 9.5 m clear height, 0.47 mm Galvalume roof and wall sheeting, a 3 m brick-wall height and three canopy zones. The project particulars and member families are summarized in Chapter 2.

1.3 Location and Site Details

The supplied project input sheet identifies Chennai as the location. The wind and earthquake inputs adopted in the training records are reproduced in Chapter 3. A precise site address, site survey and geotechnical report were not supplied; soil bearing capacity, foundation dimensions and other site-specific information are therefore not assumed.

1.4 Type of Structure

The primary system consists of steel columns and rafters forming moment-resisting transverse frames. Longitudinal hollow-section members and roof/wall bracing provide the longitudinal load path. Purlins, girts and eave struts transfer cladding loads to the primary steel and provide restraint where designed. Column bases are modelled as pinned; they still transfer vertical force, horizontal shear and possible uplift.

Source basis: [S1] project inputs and geometry, pp.1–4; [S2] joint coordinates, member incidences and support definitions. References are listed on page 24.

CHAPTER 1: INTRODUCTION (CONTINUED)

1.5 Objectives and Scope of the Training

The objectives are to understand the structural arrangement of a PEB; trace gravity and lateral load paths; calculate the design loads adopted in the source records; interpret analysis forces and displacements; explain proportioning and design checks for the main structural members; and prepare coherent design notes and study drawings. The scope covers the four weekly topics requested for the training report.

The study addresses columns, rafters, canopy beams, longitudinal struts, purlins, girts, rods and their connection requirements. Member checks available in the ANL are reported with their limitations. Components for which complete design information is unavailable are discussed through design procedures or clearly identified examples, rather than assigned unsupported sizes or pass results.

1.6 Methodology

The work follows six stages: (i) review the supplied training PDF and analysis output; (ii) extract project geometry, materials, member sizes and loading assumptions; (iii) reproduce representative load calculations; (iv) examine the model, forces, displacement summaries, code checks and warnings; (v) explain the design of primary and secondary members using source results and worked examples; and (vi) prepare a member schedule, arrangement drawings and connection concepts.

The numerical checks in this report are documentary and arithmetic checks of the supplied records. They are not a new structural analysis. In particular, an input correction requires a revised analysis before its effects on forces, member utilization or foundation reactions can be established.

1.7 Organization of the Report

Chapter 2 describes the project information and structural systems. Chapter 3 presents the first-week load calculations and combination register. Chapter 4 covers the second-week structural analysis and results. Chapter 5 explains third-week member proportioning and design. Chapter 6 presents fourth-week detailing notes and drawings. Chapter 7 concludes the study and provides the source register. The table of contents gives the page locations of individual sections.

1.8 Documentation and Evidence Basis

[S1] denotes the supplied file 1000016606.pdf, with references to its physical PDF pages. [S2] denotes R1_PEB.ANL, with references to numbered input commands or printed STAAD pages. [R1–R4] denote the external technical references listed in Chapter 7. “Calculated” identifies arithmetic from stated inputs; “illustrative” identifies a teaching example; and “recorded” identifies a result transcribed from the supplied analysis.

The cover uses demonstration student, institution, company and supervisor details, as requested. These entries do not certify attendance or site visits. The report retains the source’s design findings, including the failed checks and unresolved warnings, and the drawing sheets are marked for training use.

CHAPTER 2: DETAILED PROJECT DESCRIPTION

2.1 Preliminary Information and Structural Systems

A PEB combines engineered primary frames with repetitive secondary members and profiled cladding. Built-up tapered sections allow the depth to vary with the force pattern. Shop-fabricated components are assembled on site, commonly using bolted field connections. The selected structural arrangement must also accommodate openings, drainage, services and erection stability.

Project parameter	Record adopted for this study
Project / location	Proposed PEB industrial building / Chennai [S1, p.1]
Building plan	36.0 m transverse × 72.2 m longitudinal; plan area = 2,599.2 m²
Frames and bays	10 frame lines; 9 bays = 8.1 + 7 × 8.0 + 8.1 m
Vertical geometry	Analytical eave 10.0 m; ridge 11.5 m; stated clear height 9.5 m
Roof geometry	Two 18 m horizontal half-spans; slope 1:12; half-rafter length 18.062 m
Supports and gables	Ridge columns at interior frames; gable posts at 6 m spacing
Cladding / openings	0.47 mm Galvalume roof and walls; 3 m brick-wall height; 3 framed openings
Canopies / mezzanine	Three canopy zones with 3 m projection; no mezzanine stated
Materials in analysis	Primary steel fy/fu = 345/490 MPa; SHS 310/450 MPa; rods 250/420 MPa

2.1.1 Types of structures considered during training

System	Structural action and relevance
RCC frame	Concrete beams and columns resist bending and axial force; a useful comparison for stiffness, weight and construction sequence.
Conventional steel frame	Rolled sections or trusses; suitable for varied layouts, with member sizes selected from standard products.
PEB rigid frame	Welded tapered columns and rafters; moment joints provide transverse frame action. This is the principal system studied.
Braced steel system	Roof and wall diagonals transfer longitudinal actions; the supplied model uses tension-only rods.
Secondary / cladding system	Purlins, girts and eave struts transfer surface pressures to primary steel and provide designed restraints.

Roof rise = 18 / 12 = 1.50 m; roof angle = $\tan^{-1}(1/12) = 4.764°$; sloping length = $\sqrt{(18^2 + 1.5^2)} = 18.062$ m.

The overall width is 36 m, but the model has ridge columns. It must therefore be described as a supported two-span arrangement, rather than a 36 m unobstructed clear span. Clear architectural height and analytical node elevation are different dimensions.

Source basis: [S1] pp.1–4; [S2] joint coordinates, commands 8–45, and material/design assignments.

CHAPTER 3: STRUCTURAL ENGINEERING TRAINING – 1st WEEK

3.1 Dead Load and Imposed Load Calculations

3.1.1 Dead load (DL)

The model generates steel self-weight in global −Y and multiplies it by 1.15 to include the stated connection allowance. Roof and wall loads are added separately. The training slides use the rounded conversion 100 kg ≈ 1 kN; the following reproduction retains their adopted load values. Detailed design should use a consistent unit-weight convention.

Roof load component	Adopted value	Calculation / interpretation
Roof sheeting	5 kg/m²	0.47 mm sheet, as assumed in the source
Sag rods and flange braces	5 kg/m²	Allowance stated in the training PDF
Collateral load	10 kg/m²	Services allowance; specific supported equipment requires confirmation
Total surface allowance	0.20 kN/m²	20 kg/m² using the source conversion
Load on main-frame rafter	1.620 kN/m	0.20 × 8.1 tributary width
Purlin equivalent line load	0.551 kN/m	14 × 8.1 × 8.77 / 18.062 = 55.06 kg/m
Total applied roof DL	2.171 kN/m	1.620 + 0.551; end-frame value ≈ 1.086 kN/m

The purlin assumption is Z 270 × 75 × 20 × 2.55 mm, weighing 8.77 kg/m. Fourteen lines along an 18.062 m slope correspond to 13 spaces, or about 1.389 m average spacing. A layout limited to 1.5 m spacing requires a ceiling operation: ceil(18.062/1.5) + 1 = 14 lines. The exact eave/ridge sharing and member layout must be reconciled.

Side-column DL = 0.10 × 8.1 + 0.608 ≈ 1.410 kN/m.
Gable-post DL = 0.10 × 6.0 + 0.380 = 0.980 kN/m.
Eave-strut + gutter point load = (10.70 + 5.89) × 8.1 / 100 = 1.344 kN.

3.1.2 Roof imposed load and gutter water

The training basis adopts 0.75 kN/m² for the low-slope roof without general access. Loads in the ANL are applied as member UDLs in global −Y. For final input, distinguish loads defined on plan area from loads per sloping member length; the source directly uses the tributary-width products below.

Main-frame roof LL = 0.75 × 8.1 = 6.075 kN/m.
End-frame roof LL = 0.5 × 6.075 = 3.0375 kN/m.
Gutter-water load = 0.25 × 0.25 × 10 × 8.1 = 5.0625 kN per interior eave node.
End-frame gutter-water load = 2.5313 kN (model rounded to 2.532 kN).

The 8.1 m loading width is used in the supplied calculations even though most geometrical bays are 8.0 m. Brick-wall support, maintenance point loads, drainage blockage, erection loads and any equipment loads require an explicit design basis; they are not assumed to be verified by these calculations.

Source basis: [S1] pp.5, 9–12; [S2] commands 164–184. Imposed-load scope: [R2].

3.2 WIND AND EARTHQUAKE LOAD CALCULATIONS

3.2.1 Wind calculation reproduced from the training PDF

Input	Adopted source value
Basic wind speed and modifiers	Vb = 50 m/s; k1 = 1.0; k2 = 1.0; k3 = 1.0; k4 = 1.15
Exposure / pressure modifiers	Terrain category 2; Kd = 1.0; Ka = 0.83; Kc = 0.90
Internal pressure	Cpi = +0.5 and −0.5; source assumes 5–20% openings
Dimensions used for coefficients	h/w = 10/36 = 0.278; L/w = 72.2/36 = 2.006

Vz = Vb k1 k2 k3 k4 = 50 × 1 × 1 × 1 × 1.15 = 57.5 m/s.
pz = 0.6 Vz² = 1,983.75 N/m² = 1.984 kN/m².
pd = Kd Ka Kc pz = 1 × 0.83 × 0.90 × 1.984 ≈ 1.482 kN/m².
Net pressure = (Cpe − Cpi) pd; line load = net pressure × tributary width.

Surface / source coefficients	Worked line load at 8.1 m width
Windward wall; Cpe = +0.7, Cpi = +0.5	(+0.7 − 0.5) × 1.482 × 8.1 = +2.40 kN/m
Windward wall; Cpe = +0.7, Cpi = −0.5	(+0.7 + 0.5) × 1.482 × 8.1 = +14.41 kN/m
Roof EF at 0°; Cpe = −0.9, Cpi = +0.5	(−0.9 − 0.5) × 1.482 × 8.1 = −16.81 kN/m
Roof GH at 0°; Cpe = −0.4, Cpi = +0.5	(−0.4 − 0.5) × 1.482 × 8.1 = −10.80 kN/m

Negative pressure denotes suction on the surface. Its sign in STAAD depends on the member local axes: the ANL uses local Y for roof wind and global X/Z for wall wind. Coefficients, openings, area reductions, local cladding zones and the applicable wind map must be confirmed against the project-adopted standard; the slide values are not independently certified here.

3.2.2 Earthquake calculation reproduced from the analysis

The source uses an equivalent-static IS 1893 basis with Z = 0.16, R = 4, I = 1.2, 5% damping and medium soil. STAAD reports T = 0.53081 s, Sa/g = 2.50 and seismic weight W = 2,181.48 kN in both horizontal directions.

Ah = (Z / 2) × (I / R) × (Sa/g) = (0.16 / 2) × (1.2 / 4) × 2.5 = 0.060.
VB = Ah W = 0.060 × 2,181.48 = 130.889 kN ≈ 130.89 kN.

The seismic weight command uses SELFWEIGHT 1.5, while the DL case uses 1.15. The intended seismic mass must be reconciled. This report preserves the supplied code basis, including the PDF's IS 1893 (Part 1):2016 reference; it does not establish the code edition or amendments required for a new 2026 approval.

Source basis: [S1] pp.13–27, 36–43; [S2] commands 142–163 and printed pp.4–5.

3.3 LOAD CASES AND COMBINATION REGISTER

Case	Physical action	Model description
1 / 2	EL X+ / EL Z+	Positive-direction equivalent-static earthquake cases
3 / 4	DL / LL	Gravity self-weight, roof/wall DL, roof LL and gutter loads
5 / 6	WL 0° +Cpi / −Cpi	Transverse wind with both internal-pressure signs
7 / 8	WL 90° +Cpi / −Cpi	Longitudinal wind with both internal-pressure signs

There are 8 primary cases, 19 strength combinations (101–119) and 13 service combinations (201–213). In the register below, W5…W8 refer to wind cases 5…8, and EX/EZ refer to cases 1/2. Combination titles are reproduced separately from the numerical input where they disagree.

Combination IDs	Expressions in the supplied model
101	1.5D + 1.5L
102, 103, 104, 105	1.2D + 1.2L + 1.2W5 / W6 / W7 / W8, respectively
106, 107	1.2D + 1.2L + 1.2EX / EZ, respectively
108, 109, 110, 111	1.5D + 1.5W5 / W6 / W7 / W8, respectively
112, 113	1.5D + 1.5EX / EZ, respectively
114, 115, 116, 117	0.9D + 1.5W5 / W6 / W7 / W8, respectively
118	Title: 0.9D + 1.5EX; ACTUAL INPUT: 0.9D + 11.5EX
119	0.9D + 1.5EZ
201	D + L
202, 203	D + 0.8L + 0.8EX / EZ, respectively
204, 205, 206, 207	D + 0.8L + 0.8W5 / W6 / W7 / W8, respectively
208, 209	D + EX / EZ, respectively
210, 211, 212, 213	D + W5 / W6 / W7 / W8, respectively

3.3.1 Input checks identified during the study

LC118: command 274 is “3 0.9 1 11.5”. The earthquake factor is 7.667 times the 1.5 stated in the title. A revised factor must be confirmed and analysed; the report does not silently substitute a corrected result. **Load classification:** cases 3–8 are also tagged LOADTYPE SEISMIC, despite their DL/LL/WL titles. These labels should be corrected for clear future processing.

Direction and wind-zone coverage: explicit negative earthquake directions are absent from the listed cases. The 90° roof-pressure zones in PDF pp.37/43 also need reconciliation with ANL cases 7/8: case 8 applies +10.8 kN/m to both roof slopes, whereas the PDF table lists different zone values. Review the full required lateral-action set and local-axis signs before re-analysis.

Source basis: [S2] commands 156–302; strength/service envelopes at commands 304–307. [S1] pp.34–37, 43.

CHAPTER 4: STRUCTURAL ENGINEERING TRAINING – 2nd WEEK

4.1 Structural Model and Analysis Procedure

The model is a three-dimensional STAAD SPACE frame with 180 joints, 314 beam members and 38 pinned supports. Global X is across the 36 m width, Y is vertical and Z is along the 72.2 m length. Pins restrain translations and release support rotations; the bases still transfer vertical force, horizontal shear and possible uplift.

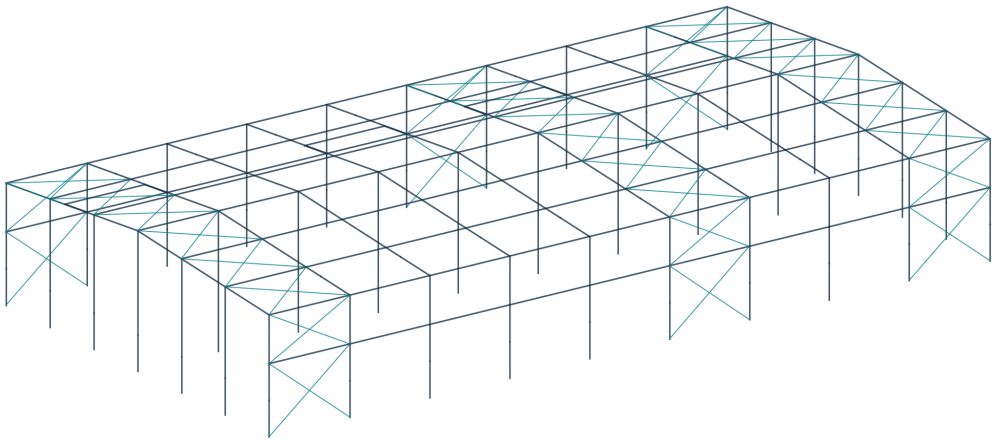


Figure 1. Analytical geometry reconstructed from the supplied joint coordinates and member incidences. Teal lines indicate the modelled rod bracing.

Analysis feature	Meaning for structural behaviour
Rigid primary frame joints	Rafters and columns transfer bending moment, shear and axial force in the transverse plane.
Longitudinal members	SHS ties and roof/wall bracing provide a longitudinal load path and restraint where detailed.
Tension-only diagonals	Members 230–289 are active in tension; the active set can change with load direction.
Member end releases	Specified SHS ends release MY/MZ; upper gable posts have end releases. Releases must match actual connection behaviour.
Rotated gable sections	BETA 90 is assigned to gable-post groups; the local major/minor axes must be checked.

4.1.1 Analysis sequence and interpretation

The study sequence is: establish geometry and connectivity; assign material and section properties; define supports, releases and restraints; apply loads; solve; check equilibrium and deformation; extract member forces; and complete strength and serviceability checks. Gravity flows from sheeting to purlins, rafters, columns and foundations. Longitudinal wind flows through the roof/wall bracing system to restrained bases.

The supplied run uses PERFORM ANALYSIS and shows tension/compression iterations for primary cases. It then uses LOAD COMB to combine results. Bentley explains that tension-only behaviour and second-order effects require simultaneous combined loading, normally using appropriate REPEAT LOAD cases, rather than superposition of separate results [R3]. Re-analysis is therefore a necessary follow-up.

Source basis: [S2] printed pp.1–8 and commands 129–163, 303; [R3].

4.2 SHEAR FORCE AND BENDING MOMENT DIAGRAMS

Shear force describes the transverse force transmitted across a cut; bending moment describes the internal couple. For a downward UDL w and the usual sagging-positive beam convention, dV/dx = −w and dM/dx = V. Thus a constant UDL gives a straight shear diagram and a parabolic moment diagram. Diagram signs must be distinguished from STAAD local member-end sign conventions.

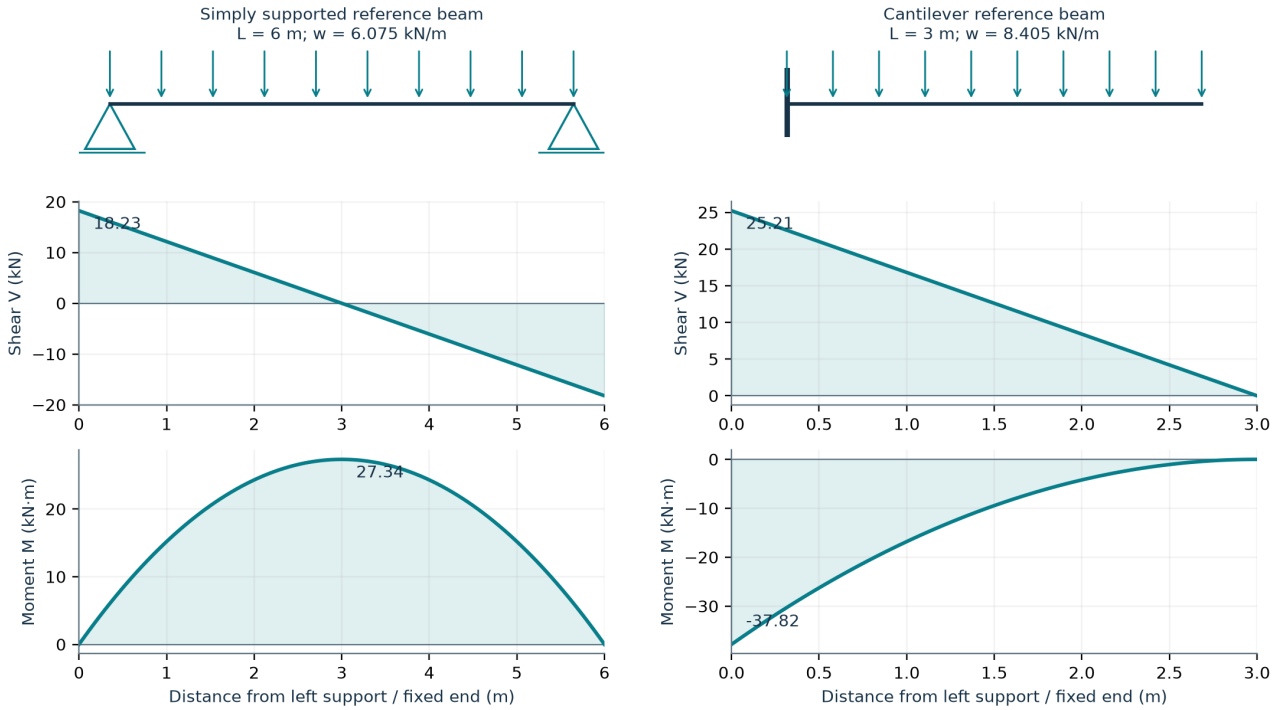


Figure 2. Calculated teaching diagrams for two idealized members. These are not extracted project-frame envelopes. The cantilever uses a downward reference load; an upward load reverses the diagram signs.

4.2.1 Example A — simply supported reference beam

Assume L = 6.0 m and w = 6.075 kN/m (the source roof-LL line-load magnitude).
RA = RB = wL/2 = 18.225 kN.
V(x) = 18.225 − 6.075x kN.
M(x) = 18.225x − 3.0375x² kN·m.
At x = L/2 = 3.0 m: V = 0 and Mmax = wL²/8 = 27.3375 kN·m.

4.2.2 Example B — canopy reference cantilever

Assume L = 3.0 m and |w| = 8.405 kN/m, matching a supplied canopy wind-load magnitude.
|Vfixed| = wL = 25.215 kN; |Mfixed| = wL²/2 = 37.8225 kN·m.

The actual canopy member has a sloping length of about 3.010 m and participates in the three-dimensional frame. Its recorded governing forces therefore differ from this isolated, unfactored reference. Portal rafters also carry axial force and end moments; the simple-beam expression wL²/8 must not replace their frame analysis. Tension-only braces principally transmit axial force, while columns resist combined axial load and biaxial bending.

Source basis: Calculated examples using stated idealizations; load magnitudes from [S2] commands 180 and 197. Actual canopy results appear on p.13.

4.3 RECORDED FORCES IN REPRESENTATIVE MEMBERS

The following values are the **concurrent critical design forces** printed at each member's governing code-check location. They are not separate maximum envelopes. FX is axial force; FY and FZ are local shears; MZ and MY are the corresponding bending moments. The dominant FY/MZ pair is shown here. Forces are in kN and moments in kN·m; negative FX denotes tension in these output rows.

Member / role	LC / x (m)	FX	FY	MZ	Ratio
21 / exterior column base segment	114 / 3.00	−222.398	69.792	−209.375	0.499
194 / upper exterior column	114 / 4.00	−203.925	47.744	−672.842	0.809
23 / eave rafter taper	114 / 0.00	−50.565	−204.161	−673.600	0.944
39 / ridge-side rafter	116 / 0.00	−148.547	−201.748	−781.111	0.985
154 / interior ridge column	118 / 11.50	61.030	95.754	−1,101*	1.039
161 / interior ridge column	118 / 11.50	61.154	95.707	−1,101*	1.038
18 / longitudinal SHS	111 / 4.05	104.595	0.000	−2.494	0.598
290 / canopy beam	114 / 0.00	−0.346	−33.809	−50.464	0.712

*The source prints −1.101E+03, so further moment precision is unavailable. x is measured from the individual member's start node, not necessarily from the building base or eave. Member 18 also has torsion MX = −0.302 kN·m. Result coordinates and signs must be retained when extracting joint design actions.

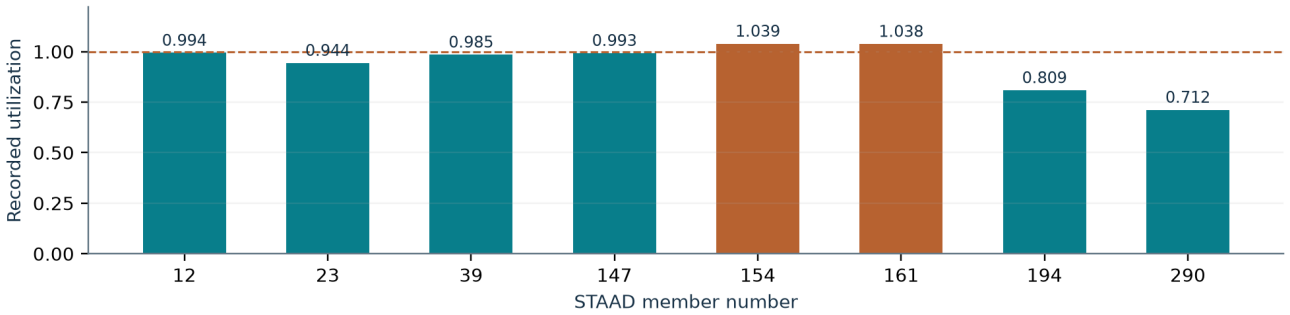


Figure 3. Selected utilization ratios as recorded in the supplied run. The dashed line is the 1.0 check threshold; input and analysis issues remain unresolved.

4.3.1 Engineering interpretation

The eave region carries substantial frame moment and shear, which explains the increased column/rafter depths near the joint. Ridge-side rafter 39 is close to its recorded shear resistance. The ridge columns reverse between compression and tension across the load set and also develop large moments. The canopy behaves as a cantilever connected to the main column system.

The reported PASS of the longitudinal SHS member is accompanied by a second-order-analysis warning. The failed ridge-column checks are governed by LC118, whose input factor is inconsistent with its title. A revised analysis may change both forces and utilization; no revised pass/fail outcome is inferred here.

Source basis: [S2] printed pp.93–100, 103–106, 135–136, 365–366, 379–380, 445–446 and 517–518.

4.4 DISPLACEMENTS, EQUILIBRIUM AND MODEL REVIEW

4.4.1 Primary-case displacement summary

The ANL reports translations in centimetres; they are converted below to millimetres. Each cell gives the signed reported extreme followed by its node number. Different component extremes generally occur at different nodes, so these entries cannot be combined as one displacement vector or directly added to form a service combination.

Case	X (mm) / node	Y (mm) / node	Z (mm) / node
3 — DL	−2.136 / 156	−6.725 / 72	−0.070 / 164
4 — LL	−3.921 / 154	−12.156 / 72	−0.094 / 164
5 — WL 0° +Cpi	+41.692 / 27	+32.669 / 44	−34.070 / 118
6 — WL 0° −Cpi	+39.442 / 3	+16.456 / 25	−3.109 / 118
7 — WL 90° +Cpi	+4.599 / 154	+27.387 / 54	+23.178 / 141
8 — WL 90° −Cpi	+6.556 / 154	+21.604 / 54	+42.808 / 118

4.4.2 Serviceability interpretation

The PDF separately illustrates 52.3 mm lateral movement against 10,000/150 = 66.7 mm, and 9.844 mm vertical rafter movement against 18,000/180 = 100 mm [S1, p.57]. Those values are not established as service envelopes of this ANL run. They are retained only as training-slide examples. Actual limits depend on cladding, supported finishes, use, connection details and the adopted design specification.

For final checking, obtain nodal and member displacements for each service combination, including the relative vertical deflection of rafters between their moving supports. Check purlins/girts, canopy tips, frame sway, gutter falls and drainage separately. A primary-case joint displacement is not a complete member-deflection check.

4.4.3 Equilibrium and warnings

DL case 3: total applied FY = −1,910.11 kN; total reaction FY = +1,910.11 kN.
Wind case 8: applied (FX,FY,FZ) = (−8.10, +3,596.40, +330.69) kN;
reaction = (+8.10, −3,596.40, −330.69) kN.

These aggregate balances support force equilibrium for the printed cases. They do not validate the assumed loads, member restraints or nonlinear combinations. Individual foundation reaction envelopes are still required, including coincident shear and uplift.

Observed issue	Required follow-up
Members 18 and 19	Moment-amplification warning above 1.4 in LC111; perform the required second-order review.
Members 154–161	Minimum web-thickness serviceability warning; investigate web slenderness and stiffening/design changes.
LC118 and tension-only bracing	Resolve the 11.5 factor and use combined-load analysis appropriate to changing brace activity.
Load definition and restraint	Verify seismic mass, wind zones/signs, lateral directions, effective lengths and actual restraint details.

Source basis: [S2] primary-case output, printed warnings and load commands; [S1] pp.57–59; combination treatment [R3].

CHAPTER 5: STRUCTURAL ENGINEERING TRAINING – 3rd WEEK

5.1 Proportioning and Member-Size Schedule

Proportioning starts from the force path, frame geometry and restraint conditions. Deeper sections are placed where bending demand is high; flange area resists bending/axial action, while the web transfers shear and stabilizes the flanges. Economy depends on plate thickness, fabrication repetition, transport limits, local buckling and serviceability, not only on section area.

Member family / examples	Depth D, start → end (mm)	Flanges b × t (mm)	Web (mm)
Exterior column, base segment 1/21/etc.	524 → 659	250 × 12	6
Exterior column, middle 162/169/etc.	659 → 794	250 × 12	6
Exterior column, upper 192–211	794 → 974	250 × 12	6
Ridge columns 154–161	500 → 1,150	350 × 25	6
Gable posts 5–9, 149–153, 164–168, 187–191	420 → 420	200 × 10	6
Main eave rafters 23/27/38/42/etc.	924 → 524	275 × 12	6
Main constant-depth rafters 26/28/etc.; 299–314	524 → 524	275 × 12	5
Main ridge-side rafters 24/25/39/40/etc.	974 → 524	275 × 12	6
End-frame eave rafters 3/12/143/147	524 → 424	200 × 12	5
End-frame other rafters 4/10/11/13/144–146/148	424 → 424	200 × 12	5
Canopy beams 290/291/293–296	316 → 316	150 × 8	5
Longitudinal ties / struts	SHS 150 × 150 × 4	Closed hollow section	4
Braces 230–289	Solid circular rod Ø25	Tension-only specification	—
Canopy Z members 292/297/298	260ZS75X2 (source designation)	Cold-formed section	2 nominal

5.1.1 Reading the tapered-section input

The seven TAPERED values are Dstart, tw, Dend, btop, ttop, bbottom and tbottom [R4]. D is the overall section depth; clear web depth is D − ttop − tbottom. Equal flanges are assigned in the listed families. Start/end follow the member incidence order, so the schedule must not be reversed during fabrication interpretation.

For a symmetric I section: $A = 2btf + (D - 2tf)tw$.
For D = 524, b = 275, tf = 12, tw = 5 mm: A = 9,100 mm² = 91.0 cm².

This is a schedule of analytical section assignments, not an approved fabrication bill. Original PDF elevation notes and the ANL are not assumed to be the same revision. Main-roof purlins are represented through applied loads; their sizes and checks need the separate secondary-member design described on p.18.

Source basis: [S2] commands 105–128; parameter interpretation [R4].

5.2 DESIGN OF COLUMNS AND GABLE POSTS

5.2.1 Column design procedure

For each column, identify the effective length and restraint about both axes; classify flange and web plates; determine compression resistance; check tension reversal where applicable; evaluate bending, shear and axial–moment interaction; and check deflection and connection demands. A low axial-force ratio alone does not establish adequacy under frame bending.

Compression design framework: $P_d = A_e f_{cd}$.
Elastic reference stress: $f_{cc} = \pi^2 E / (KL/r)^2$.
For a prismatic compression member: $\bar{\lambda} = \sqrt{(f_y/f_{cc})}$;
$\varphi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/[\varphi + \sqrt{(\varphi^2 - \bar{\lambda}^2)}] \leq 1$;
$f_{cd} = \chi f_y/\gamma_{m0}$. Select the code buckling curve and applicable section area.

These equations explain the role of slenderness and imperfections; a tapered column requires the applicable tapered-member treatment and appropriate effective properties. E = 205,000 MPa is used in the supplied model. End restraints and intermediate restraints must be supported by actual details.

5.2.2 Worked interpretation — ridge column 154

Parameter / result	Value from the supplied check
Geometry	11.5 m long; D = 500 → 1,150 mm; flanges 350 × 25 mm; web 6 mm
Critical section area	At top: A = 2(350)(25) + (1,150 − 50)(6) = 24,100 mm²
Local slenderness indicators	Top clear web ratio = 1,100/6 = 183.3; flange outstand ratio = (350 − 6)/(2 × 25) = 6.88
Program classification	Slender web; plastic flange; overall section class slender
Effective-length inputs	LZ = LY = 11.5 m; KZ = 2.0; KY = 1.0; reported slenderness 122.28
Compression envelope	317.396 kN in LC101; printed minor-axis compression resistance 1,806 kN
Critical bending	|MZ| ≈ 1,101 kN·m in LC118; printed resistance ≈ 1,093 kN·m
Recorded governing result	Interaction ratio 1.039, FAIL; additional web-thickness serviceability warning

The 317.396 kN compression and 1,101 kN·m moment arise from different cases and must not be combined arbitrarily. At the governing LC118 station, the concurrent axial compression is 61.030 kN. Member 161 similarly records ratio 1.038. Both results must be revisited after the load-case and analysis corrections.

5.2.3 Exterior columns and gable posts

Exterior columns comprise 3 m, 3 m and 4 m analytical segments with increasing depth toward the eave. Segmentation does not itself provide lateral restraint. Gable posts use 420 mm sections rotated by BETA 90 and transfer end-wall wind to the roof/bracing and bases. Their connection releases and effective lengths must be matched to the installed arrangement.

Source basis: [S2] printed pp.365–366 and 379–380; commands 130, 321–340. Design topics: [S1] pp.45–51, 65–67.

5.3 DESIGN OF RAFTERS AND CANOPY BEAMS

5.3.1 Rafter design principles

A rafter check begins with plate classification and the compression-flange restraint under each load direction. Gravity and wind uplift can reverse the compression flange. Purlins connected to the upper flange do not automatically restrain the lower flange; flange stays and their connection/load path must be designed. Slender webs require explicit treatment of shear buckling and flexural resistance.

For a suitable laterally supported section: $M_d = \beta_b Z_p f_y / \gamma_{m0}$, subject to code limits.
For a semi-compact section $\beta_b Z_p$ corresponds to Z_e.
For an unrestrained member: M_d depends on lateral-torsional buckling strength.
Shear yielding reference: $V_d = A_v f_y / (\sqrt{3}\,\gamma_{m0})$; reduce/check for shear buckling where applicable.

These are design frameworks, not a replacement for the slender-web and high-shear rules applicable to the supplied sections. Bending resistance can reduce when shear is high, and axial–moment interaction must be checked using concurrent forces.

5.3.2 Worked interpretation — ridge-side rafter 39

Check	Demand / resistance	Interpretation
Section and restraint	D 974 → 524; b 275; tf 12; tw 6 mm	LX = LY = 1.5 m; LZ = 15.36 m as assigned
Classification at deep end	Clear web = 950 mm; d/tw = 158.3	Slender web; semi-compact flange in output
Critical local shear FY	|−201.748| / 204.891 = 0.9847	Governing ratio rounds to 0.985 in LC116
Major-axis bending	|−781.111| / 982.925 = 0.7947	Printed bending ratio 0.795; high-shear treatment cited by output
Section interaction	0.836 in LC116	Separate from the governing shear ratio
Recorded status	PASS, ratio 0.985 at x = 0.0 m	Conditional on supplied inputs and unresolved model review

The program labels the FY-related check “Shear Minor” and the MZ check “Bend Major”; this report retains numerical components to avoid ambiguity. A 0.985 ratio leaves a small numerical margin in this run. Restraint spacing, load definition, fabrication tolerances and revised analysis can affect the result.

5.3.3 Canopies and end-frame rafters

Canopy beams 290, 291 and 293–296 have D = 316 mm, 150 × 8 mm flanges and 5 mm webs. Member 290 records MZ = −50.464 kN·m and FY = −33.809 kN under LC114, with ratio 0.712. The canopy connection must transfer the cantilever moment into the main column; drainage, uplift and lateral restraint require separate details.

End-frame rafters use 200 × 12 mm flanges with 424/524 mm depths. Member 12 records ratio 0.994 in LC118. Its result should be revisited after the LC118 discrepancy is resolved, rather than treated as an optimized final section.

Source basis: [S2] printed pp.81–82, 135–136 and 517–518. Design concepts: [S1] pp.61–68.

5.4 SECONDARY STEEL AND BRACING DESIGN

Component	Section / evidence	Required design checks
Main-roof purlin	PDF assumption: Z 270 × 75 × 20 × 2.55 mm; 8.77 kg/m	Cold-formed effective section, biaxial bending, local/distortional buckling, uplift and deflection
Side-wall girt	PDF: same Z section for 8.1 m bay	Wind pressure/suction, continuity, cleats, sheeting restraint and deflection
Gable-wall girt	PDF: Z 230 × 75 × 20 × 2 mm; 6.32 kg/m	6 m bay bending, local buckling, fasteners and end connections
Eave strut	PDF: CS 270 × 75 × 20 × 3.15 mm; 10.7 kg/m	Roof/wall reactions, gutter action, biaxial bending and axial action
Longitudinal SHS	ANL: 150 × 150 × 4 mm	Compression, tension, bending, slenderness, connections and second-order behaviour
Canopy secondary members	ANL 292/297/298: 260ZS75X2	Separate cold-formed check; omitted from listed CHECK CODE members
Roof / wall bracing	ANL 230–289: Ø25 solid rods	Gross yielding, threaded/net-section rupture, connections, slack and load reversal

5.4.1 Illustrative roof-purlin calculation

To demonstrate the method, idealize one 8.1 m purlin span as simply supported with 1.5 m tributary roof width. Use the source DL allowance of 0.20 kN/m² plus its assumed purlin self-weight. This is a service-load example; actual lap continuity, roof slope and restraint need a cold-formed design.

$w_{DL} \approx 0.20 \times 1.50 + 0.0877 = 0.3877$ kN/m.
$w_{LL} = 0.75 \times 1.50 = 1.1250$ kN/m; $w_{total} = 1.5127$ kN/m.
$R = wL/2 = 6.126$ kN; $M_{max} = wL^2/8 = 12.406$ kN·m.
Using the source's illustrative L/240 limit: $\delta_{allow} = 8{,}100/240 = 33.75$ mm.
$I_{required} = 5wL^4/(384E\delta_{allow}) \approx 12.25 \times 10^6$ mm⁴ (E = 205,000 MPa).

The inertia requirement is an idealized strong-axis stiffness demand, not a pass result for the named Z section. Effective stiffness, weak-axis action, continuity and manufacturer properties must be checked. IS 800's scope identifies cold-formed light-gauge sections as requiring separate provisions [R1].

5.4.2 Illustrative gross-yielding resistance of a Ø25 brace

$A_g = \pi \times 25^2/4 = 490.87$ mm².
$T_{dg} = A_g f_y/\gamma_{m0} = 490.87 \times 250/1.10 = 111.56$ kN.

This is only the gross-section yielding resistance. Threaded net area, rupture, gusset block shear, rod-end/turnbuckle resistance and actual factored axial demand can govern. The ANL assigns rod material values but excludes members 230–289 from its CHECK CODE list; their final adequacy is not established.

Source basis: [S1] pp.5, 10–11 and 59; [S2] commands 121–128, 132–133, 311–312 and 351; [R1].

5.5 DESIGN COVERAGE, CONNECTIONS AND QUANTITIES

5.5.1 Coverage of the supplied design output

The CHECK CODE command covers members 1–229, 290–291, 293–296 and 299–314: 251 members. Counting the status lines gives 249 PASS and 2 FAIL results. The other 63 members comprise 60 tension-only rods and three canopy Z members. Presence in the analytical model does not demonstrate that a member or connection has been designed.

Design item	Evidence available	Status for this report
Primary columns and rafters	Sizes, loads and detailed member code checks	Recorded results summarized; correction/re-analysis required
SHS ties / struts	Member checks included	Second-order warnings for members 18/19 remain open
Rods / cold-formed members	Rod sizes; limited Z assignments; PDF assumptions	Separate member and connection checks needed
Knee, ridge and splice joints	No complete connection calculations supplied	Design workflow and concept details provided
Base plates / anchors / foundations	Pinned-base model; aggregate equilibrium output	Individual reaction envelopes, anchor and geotechnical design needed

5.5.2 Connection design method

Extract the concurrent N, V and M at each joint for every governing case, including reversal. Select a connection arrangement that provides the stiffness/release assumed in the model. Check bolts in shear, tension and interaction; bearing, net section and block shear of connected plates; end-plate bending and prying; weld strength; and column/rafter local effects. An initial moment-couple estimate is $T \approx M/z$, but actual bolt-row distribution and prying must be analysed.

For a pinned base, determine compression bearing, uplift anchorage and shear transfer consistently with the assumed rotational release. Foundation design then checks soil pressure, sliding, overturning, uplift, concrete bending, shear and punching. Site soil data, concrete grade, embedment, anchor grade and base reactions are not supplied; foundation dimensions are therefore not invented.

5.5.3 Steel quantity reported by the model

Total reported model steel weight = 775.328 kN.
Equivalent mass using g = 9.81 m/s² ≈ 79.03 tonnes.
Model mass / plan area ≈ 79,034 / 2,599.2 = 30.41 kg/m².

This is the weight of the members represented in the model. Main-roof purlins and much of the girt/eave-strut system are accounted for as applied loading rather than explicit steel members. Connections, base plates, stiffeners, coating, sheeting and procurement waste are not established by this take-off. The 15% self-weight load factor is not a measured connection quantity.

Week 3 deliverable: a traceable member schedule and explanation of design checks across all structural families. The source is insufficient for a complete approved building design; missing checks are identified rather than replaced with invented pass results.

Source basis: [S2] printed p.58, steel take-off; commands 349–351. Connection design study basis: [S1] pp.54–56 and 67.

CHAPTER 6: STRUCTURAL ENGINEERING TRAINING – 4th WEEK

6.1 General Notes for Design and Detailing

The following notes form a training specification to accompany the study drawings. They translate analysis assumptions into drawing requirements. Final material grades, tolerances, connection sizes and inspection criteria must be assigned by the responsible designer and project specification.

Topic	General design / detailing note
Units and dimensions	All drawing dimensions are in millimetres unless stated. Levels relate to the model base datum. Do not scale drawings. Verify site dimensions and openings before fabrication.
Material identification	Primary member design values are fy = 345 MPa and fu = 490 MPa; SHS and rod values differ. Specify the actual product grade, thickness-dependent strength and certificates in procurement documents.
Built-up members	Show flange/web thicknesses, taper direction, plate joints, weld symbols and stiffeners. Analytical division points do not automatically represent physical splices.
Connections	Specify bolt grade, diameter, hole type, spacing, edge distances and tightening method. Show end plates, cleats, gussets, base plates and anchor projections with design references.
Welds and fabrication	Show weld type, size, effective length and required continuity. Use approved welding procedures; inspect as required by the project quality plan. Do not prescribe unverified weld sizes from this report.
Lateral restraint	Locate purlin/girt attachments, flange stays, sag rods and braces explicitly. Verify the compression-flange restraint under gravity and uplift. A 1.5 m software length requires a corresponding physical restraint system.
Bracing and erection	Identify permanent braced bays and connections. Provide a planned erection sequence and temporary stability system; do not assume partially assembled frames are stable.
Base and civil interface	Coordinate anchors, pedestal levels, grout, base rotation assumptions and foundation reactions. Model pins do not remove uplift or horizontal shear demand.
Cladding and drainage	Coordinate sheet laps, screws, flashings, gutters and downpipes. Check local suction zones, openings and water accumulation; prevent incompatible restraint at expansion interfaces.
Durability and maintenance	Define preparation, coating/galvanizing and touch-up requirements for the exposure. Avoid water traps and inaccessible unprotected joints; coordinate fire protection where required.

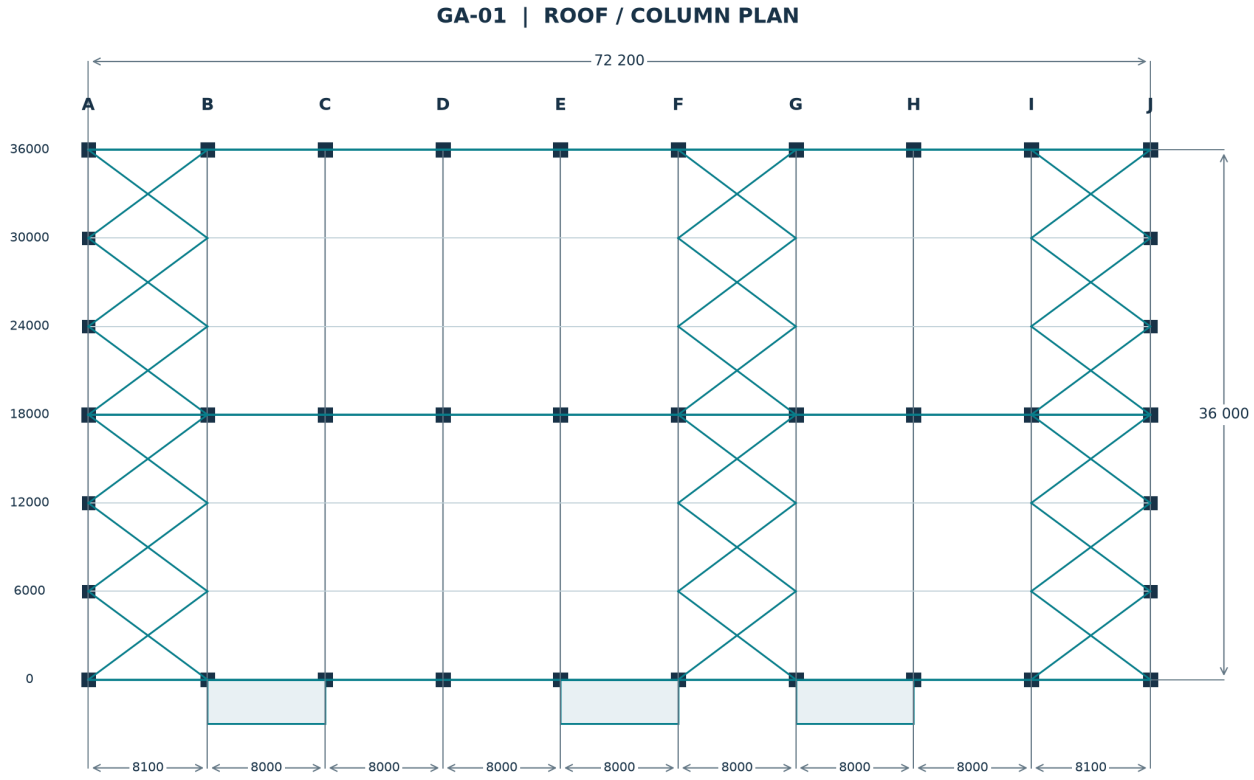
6.1.1 Drawing register

Sheet	Report page	Contents / issue status
GA-01 / GA-02	21	Column/roof arrangement and transverse elevation — training geometry
D-01 to D-04	22	Primary profiles and secondary sections — analytical sizes
C-01 to C-04	23	Base, knee, ridge and bracing/cleat connections — schematic design intent

No approved fabrication drawings were supplied. Pages 21–23 are newly prepared study drawings. They include source dimensions where available and retain design-dependent connection dimensions as variables.

Source basis: Drawing requirements developed from [S1]/[S2] and the general design/fabrication scope in [R1].

6.2 GENERAL ARRANGEMENT AND FRAME ELEVATION



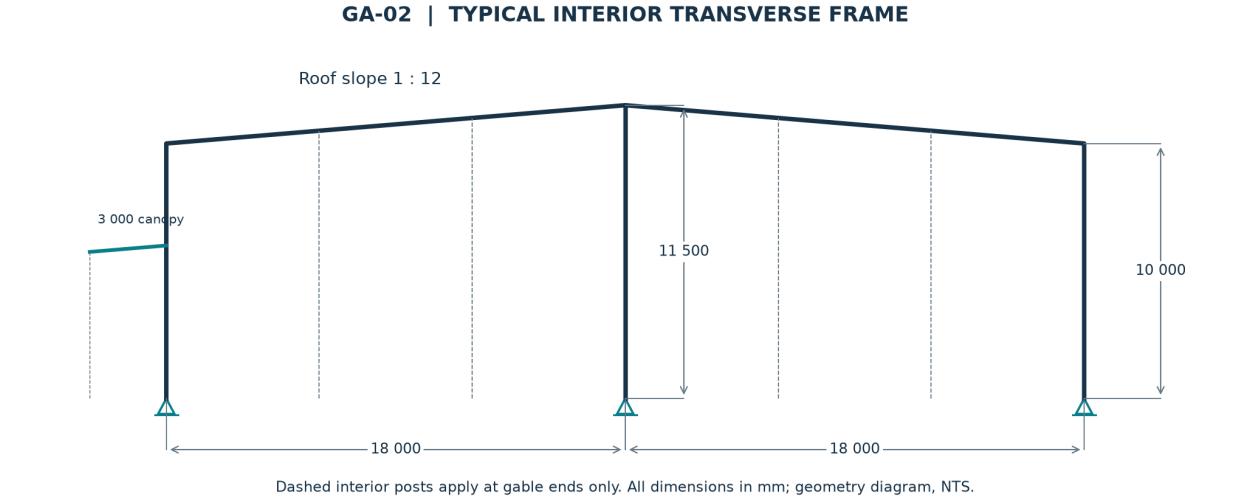


Figure 4. Dimensioned arrangement reconstructed from the ANL. Roof bracing is shown in the first, central and last braced bays. Three canopy zones are on the X = 0 side. NTS; dimensions in mm.

Main column lines are at X = 0, 18,000 and 36,000. Interior ridge columns occur at Z = 8,100 through 64,100. Gable-end posts fill the 6,000 mm transverse intervals. Side-wall bracing is also modelled in the corresponding braced bays, with wall segmentation at 6,000 mm height. Column intermediate nodes occur at 3,000 and 6,000 mm.

The 9,500 mm clear-height requirement is architectural; verify clearance below haunches and services separately from the 10,000 mm analytical eave node. Model centre lines do not define cladding faces, foundations or fabrication offsets.

Source basis: [S2] commands 8–94. Frame grid letters are report labels introduced for readability.

6.3 MEMBER PROFILES AND SECTION DETAILS

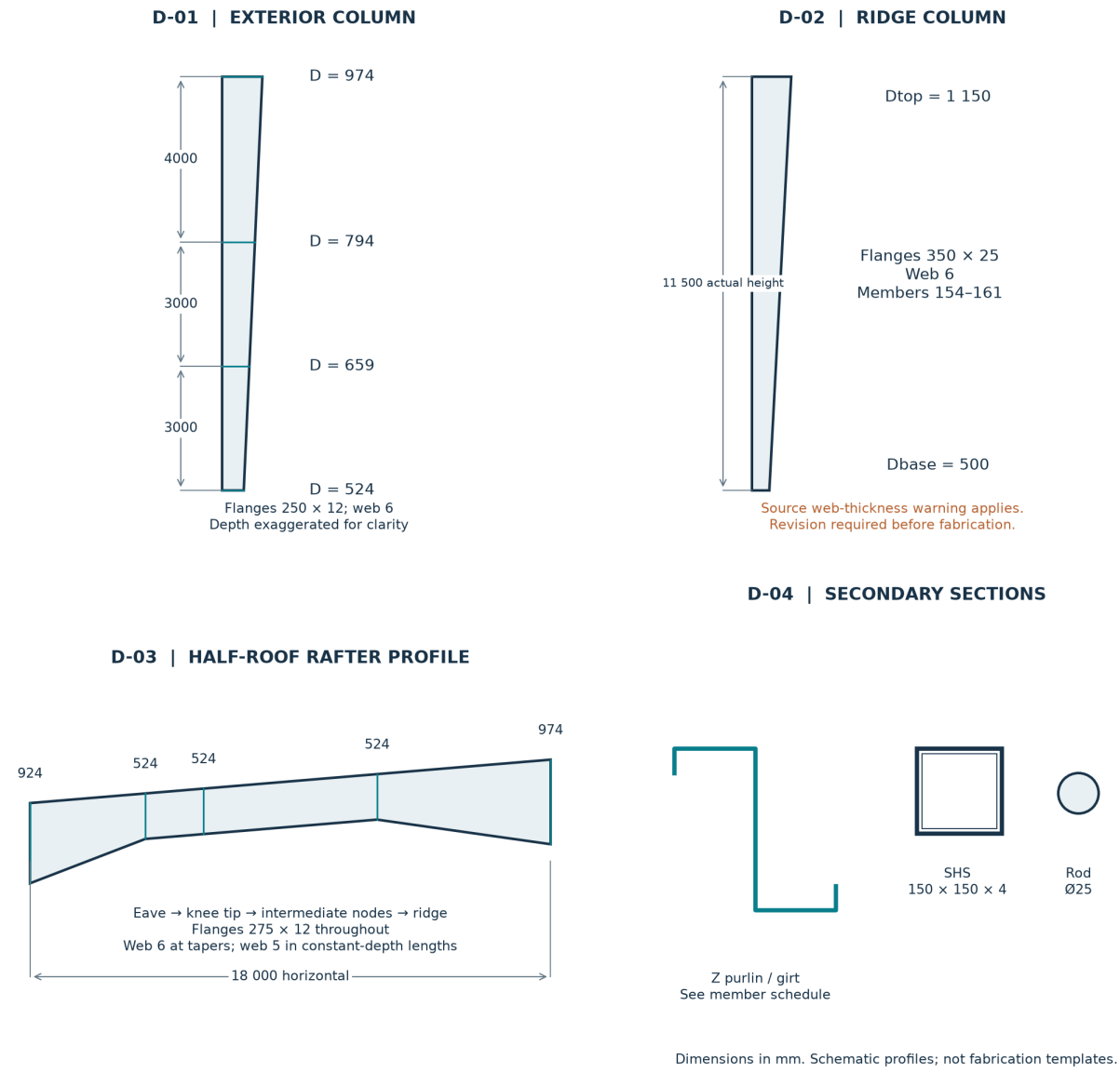


Figure 5. Primary-member profiles and secondary-section concepts. Taper direction follows the representative model member orientation; section depths are exaggerated where needed for legibility. Secondary-member sizes are listed on page 18.

Additional member detail	Size / detailing action
Gable posts	D 420; b 200; tf 10; tw 6 mm. Show BETA-90 section orientation and the released roof connection.
End-frame rafters	D 524 → 424 or constant 424; b 200; tf 12; tw 5 mm. Confirm piece marks and splice positions.
Canopy beams	D 316; b 150; tf 8; tw 5 mm. Projection 3,000; fall 250; check cantilever root connection.
Secondary-member details	Main purlin/girt assumptions are on p.18. Show actual lip dimensions, laps, cleats, sag rods and flange stays after design.

Dimension key: D = overall depth; b = flange width; tf = flange thickness; tw = web thickness. Final fabrication drawings must add plate lengths, bevels, welds, stiffeners, holes and approved connection dimensions.

Source basis: [S2] member-property commands 105–128; [S1] secondary-member assumptions, pp.5 and 10–11.

6.4 CONNECTION DETAILS AND DRAWING COMPLETION

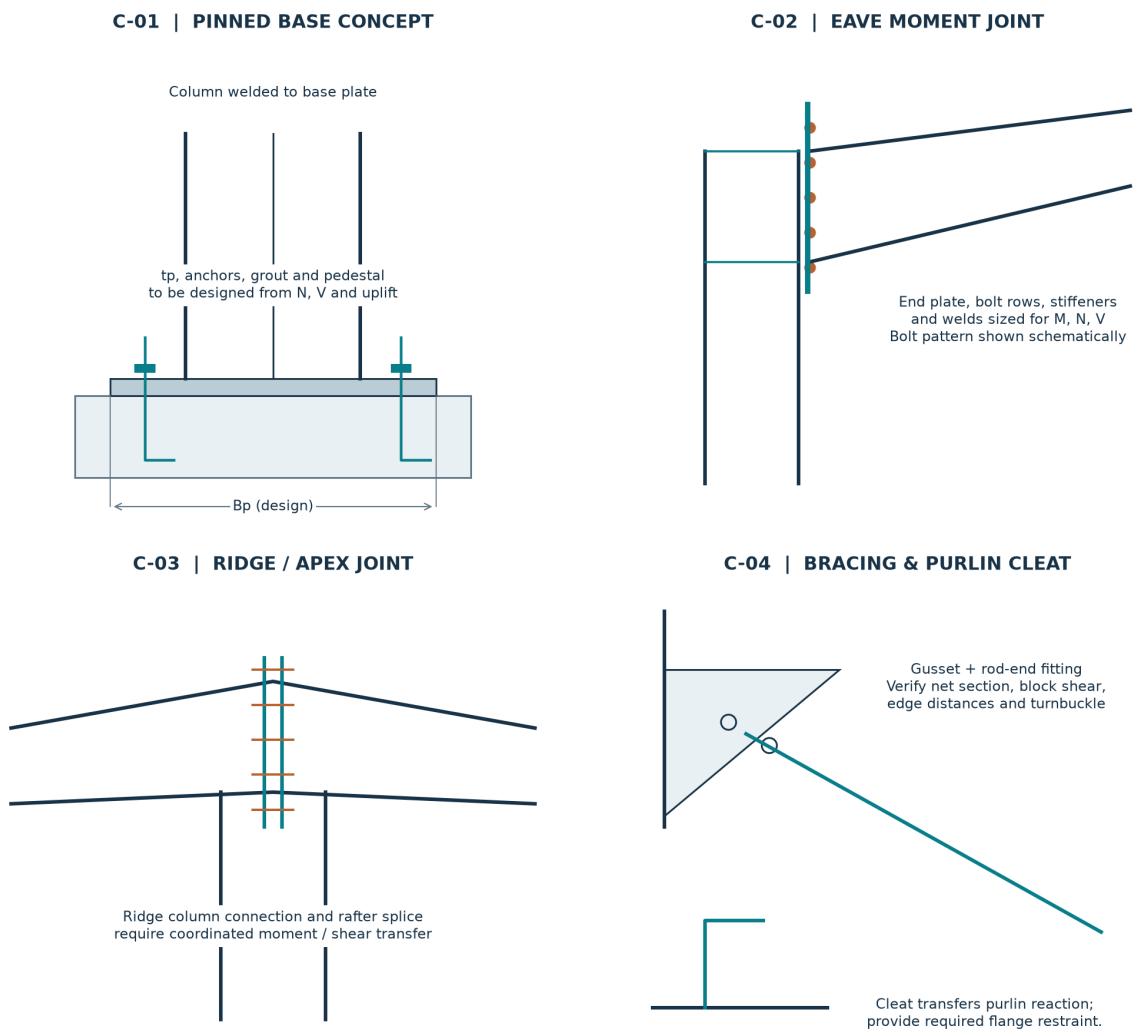


Figure 6. Connection concepts illustrate load transfer and detailing requirements. Bolt symbols are schematic: the number, diameter and spacing are not designed. No detail is issued for construction.

Detail	Actions to obtain	Items required before issue
C-01 Base	Coincident compression/uplift and shear	Bp, Lp, tp; anchor grade/diameter/embedment; grout and pedestal design
C-02 Eave joint	Rafter/column end N, V and M, both signs	Bolt-row forces; prying; plate/weld/stiffener design; erection access
C-03 Ridge	Both rafter ends and ridge-column actions	Joint stiffness and splice location; bolt/weld group checks; node eccentricity
C-04 Bracing / cleat	Factored rod tension; purlin/girt reaction	Gusset and threaded area; turnbuckle rating; restraint capacity and cleat checks

Drawing completion requires a coordinated general arrangement, member piece marks, connection calculations, erection marks, anchor plan, member/bolt lists and revision history. Each change affecting stiffness, release or load path should be returned to the analytical model. The final check is agreement among the calculation model, member schedule and drawings.

Source basis: New training sketches informed by [S1]/[S2]. Connection sizes and foundation design remain open because the required calculations and input data were not supplied.

CHAPTER 7: CONCLUSION

7.1 Learning Outcomes, Conclusions and References

7.1.1 Weekly learning outcomes

Week 1: interpreted the project geometry, distinguished frame types and reproduced gravity, wind and earthquake load calculations. **Week 2:** explained the model, plotted reference shear/moment diagrams and interpreted member forces, displacement summaries and equilibrium. **Week 3:** documented section proportions and studied compression, bending, shear, bracing and secondary-member design. **Week 4:** prepared general notes, arrangement drawings, member profiles and connection design concepts.

7.1.2 Conclusion and remaining engineering work

The records support a useful study of a tapered steel PEB with ridge-supported roof frames. They also show why analysis output must be reviewed before drawing a design conclusion. The most significant findings are the LC118 factor discrepancy, inappropriate result superposition for tension-only behaviour, two failed ridge-column checks, second-order warnings, web-thickness warnings and incomplete design coverage for rods, secondary steel and connections.

The next engineering revision should confirm the adopted code editions and project loads; reconcile wind zones, seismic mass and load directions; correct and rerun the model using appropriate combined loading and second-order treatment; then complete strength/serviceability checks, connections, anchors and foundations. The report does not claim that these corrections have already been made.

7.1.3 Source register and references

[S1] User-supplied training PDF: 1000016606.pdf, 68 pages. Project and geometry pp.1–4; gravity loads pp.5, 9–12; wind/seismic pp.13–27, 36–43; analysis/design topics pp.29–35, 45–68. Screenshots were visually reviewed; no searchable text layer was present.

[S2] User-supplied STAAD.Pro output: R1_PEB.ANL. CONNECT Edition 22.00.00.15; output dated 17 October 2026; 560 printed pages. Input/model pp.1–8; design parameters pp.56–57; take-off p.58; member checks pp.59–560. Original files were not altered.

[R1] Bureau of Indian Standards — IS 800:2007, official scope preview. General steel-construction scope; loading standards and separate provisions for cold-formed sections. Detailed calculations in this report follow the supplied study records.

[R2] Bureau of Indian Standards — IS 875 (Part 2):1987, official scope preview. Scope of imposed loads for buildings. Roof-load values used here are those adopted in the supplied PDF.

[R3] Bentley Systems — Member Tension And Combination Load Cases. Explains the difference between result combinations and simultaneously applied repeat loads for nonlinear/tension-only behaviour.

[R4] Bentley Systems — Assign Tapered I Beam Property and Material using OpenSTAAD. Defines the seven tapered I-section dimensions used to interpret the ANL member-property schedule.

Standards named in the supplied study include IS 800:2007, IS 875 Parts 1 and 2:1987, IS 875 Part 3:2015 and IS 1893 Part 1:2016. These identify the source basis, not a statement that every edition remains applicable to a new approval. Web references checked 21 October 2026.